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Method of fabricating a semiconductor structure

Abstract

A method of manufacturing a semiconductor structure forming a first diffusion layer on a first electrode layer and forming a core layer over the first diffusion layer. A second diffusion layer is formed over the core layer. A plurality of diffusion regions are formed in the second diffusion layer. A second electrode layer is formed over the second diffusion layer and in contact with the plurality of diffusion regions. The second diffusion layer is coupled to the plurality of diffusion regions through the second electrode layer. The substrate is sandwiched between the first electrode layer and the second electrode layer.

Inventors: Long; Tao (Shanghai, CN), Huang; Pin-Hao (Taipei, TW), Chen; Ze Rui (Milpitas, CA)

Applicant: Diodes Incorporated (Plano, TX)

Family ID: 87314633

Assignee: Diodes Incorporated (Plano, TX)

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Primary Examiner: Rahman; Mohammad A
Attorney, Agent or Firm: AP3 Law Firm PLLC

Background/Summary

(1) This application is a Divisional of claims the benefit of and priority to U.S. patent application Ser. No. 17/690,842 filed on Mar. 9, 2022. Said application to claims the benefit of and priority to

TECHNICAL FIELD

(1) The present invention relates to a semiconductor structure and method of fabricating the same, and more particularly, to a semiconductor structure having a plurality of diffusion regions.

BACKGROUND

(2) In a flyback converter circuit, the diode in a Resistor-Capacitor-Diode (RCD) circuit usually needs to have a higher conduction speed and a larger reverse recovery charge to suppress or slow down voltage spikes and circuit oscillations caused by the leakage inductance of the transformer. If a general fast recovery diode (FRD) is used, it may not be able to turn on quickly to transfer energy to the capacitor, thereby resulting in voltage spikes and circuit oscillations, and even affecting the overall circuit efficiency.

(3) In the prior art, a bipolar junction transistor (BJT) can also be used. In particular, the base and the emitter of the BJT are short-circuited. After the base and the emitter have shorted together, the BJT is used as a diode element in the RCD circuit to improve the efficiency of the circuit. However, the manufacturing process of the BJT is complex and the cost is higher.

SUMMARY

(4) Embodiments of the present disclosure relate to a semiconductor structure. The semiconductor structure includes a substrate having a first conductivity type. The substrate has a first surface and a second surface opposite to the first surface. The semiconductor structure also includes a first diffusion layer having the first conductivity type disposed in the substrate and adjacent to the first surface of the substrate. The semiconductor structure has a first electrode layer disposed on the first diffusion layer. The semiconductor structure further includes a second diffusion layer having a second conductivity type disposed in the substrate and adjacent to the second surface. The semiconductor structure further includes a plurality of diffusion regions having the first conductivity type and disposed in the second diffusion layer. The semiconductor structure further includes a second electrode layer disposed on the second diffusion layer and in contact with the plurality of diffusion regions such that the second diffusion layer is coupled to the plurality of diffusion regions through the second electrode layer, and the substrate is between the first electrode layer and the second electrode layer.

(5) Embodiments of the present disclosure relate to a method of fabricating a semiconductor structure. The method includes providing a substrate of a first conductivity type having a first surface and a second surface opposite to the first surface. The method also includes diffusing into the substrate from the first surface of the substrate to form a first diffusion layer having the first conductivity type, and diffusing from the second surface of the substrate into the substrate to form a second diffusion layer having a second conductivity type. The method further includes forming a plurality of diffusion regions having the first conductivity type in the second diffusion layer, and forming a first electrode layer on the first diffusion layer. The method also includes forming a second electrode layer on the second diffusion layer, the second electrode layer in contact with the plurality of diffusion regions such that the second diffusion layer is coupled to the plurality of diffusion regions through the second electrode layer, and the substrate is between the first electrode layer and the second electrode layer.

(6) The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter which form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent

constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present disclosure, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 shows a cross-sectional view of a semiconductor structure 1 in accordance with various embodiments of the present disclosure;
- (3) FIG. 2 shows a cross-sectional view of a semiconductor structure 2 in accordance with various embodiments of the present disclosure;
- (4) FIG. 3A shows a top view of the semiconductor structure in accordance with various embodiments of the present disclosure;
- (5) FIG. 3B shows another top view of the semiconductor structure in accordance with various embodiments of the present disclosure;
- (6) FIG. 3C shows another top view of the semiconductor structure in accordance with various embodiments of the present disclosure; and
- (7) FIGS. 4A-4I show one or more stages in a method of fabricating the semiconductor structure in accordance with various embodiments.
- (8) Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the various embodiments and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

- (9) The making and using of the presently preferred embodiments are discussed in detail below. It should be appreciated, however, that the present disclosure provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the disclosure, and do not limit the scope of the disclosure.
- (10) The following disclosure provides many different embodiments or examples for implementing different features of the provided subject matter. Specific examples of components and configurations are described below. Of course, these are examples only and are not intended to be limiting. In this disclosure, forming a first feature on or over a second feature may encompass embodiments where the first feature is formed in direct contact with the second feature, and may also encompass embodiments where the first feature is formed not in direct contact with the second feature. Furthermore, the present disclosure may repeat reference numerals and/or letters in various embodiments. This repetition is for simplicity and clarity. The repetition, by itself, does not indicate a relationship between the various embodiments and/or configurations discussed.
- (11) Embodiments of the present disclosure are discussed in detail below. It should be appreciated, however, that the present disclosure provides many applicable concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are illustrative only, and do not limit the scope of the disclosure.
- (12) The present disclosure provides a semiconductor structure and a method of fabricating the same. Compared with a general fast recovery diode (FRD), the semiconductor structure of the present disclosure has a structure similar to the NPN structure or the PNP structure of a bipolar junction transistor (BJT). The semiconductor structure has a higher turn-on speed and a short reverse recovery time. The semiconductor structure can effectively suppress or slow down voltage spikes and circuit oscillations. In addition, compared with general bipolar junction transistors, the

method for fabricating the semiconductor structure of the present disclosure uses the double diffusion technique and the glass passivation parts technique. The method does not need to use the epitaxial growth technique, and need to use less mask layers. As a result, the process is simplified and the cost is lower.

(13) Referring to FIG. 1, FIG. 1 shows a cross-sectional view of a semiconductor structure **1** in accordance with various embodiments of the present disclosure. The semiconductor structure **1** may include a substrate **10**, electrode layers **15** and **16**, a polysilicon layer **17** and a glass layer **18**.

(14) The substrate **10** may comprise a semiconductor substrate such as a silicon wafer. In some embodiments, the thickness H1 of the substrate **10** may be in a range from about 220 micrometers (μm) to about 240 μm . In some embodiments, the semiconductor structure **1** may be formed using an N-type single crystal silicon wafer having a volume resistivity in a range from about 30 Ohm-cm to about 45 Ohm-cm.

(15) The substrate **10** may have a surface **101** and a surface **102** opposite to the surface **101**. In some embodiments, the width of the surface **101** may be greater than the width of the surface **102** when viewed in a cross-sectional view. In some embodiments, the total surface area of the surface **101** may be greater than the total surface area of the surface **102**.

(16) In some embodiments, the substrate **10** may include a diffusion layer **11** adjacent to the surface **101**, a diffusion layer **13** adjacent to the surface **102**, and a core layer **12** between the diffusion layer **11** and the diffusion layer **13**. The diffusion layer **13** can be equivalently used as a base, and the diffusion layer **11** can be equivalently used as a collector.

(17) The core layer **12** can be a portion of the original silicon wafer. The core layer **12** can be doped with N-type impurities. As a result, the region is marked as an “N” region in FIG. 1.

(18) The diffusion layer **11** may include a semiconductor layer doped with N-type impurities. The diffusion layer **11** and the core layer **12** may have the same conductivity type. However, the impurity concentration of the diffusion layer **11** may be higher than the impurity concentration of the core layer **12**. As such, the diffusion layer **11** is marked as an “N+” region in FIG. 1. In some embodiments, the diffusion layer **11** may be formed by diffusing N-type impurities into the substrate **10** from the surface **101** of the substrate **10**. The N-type impurities may include pentavalent elements such as phosphorus, arsenic and antimony. In some embodiments, a diffusion boundary **112** may be formed between the diffusion layer **11** and the core layer **12**. In some embodiments, the diffusion boundary **112** is a boundary between a high concentration N-type impurity (N+) region and an N-type impurity (N) region. The impurity concentration in an “N+” region or a “P+” region used in this disclosure is higher than that in its adjacent regions.

(19) The diffusion layer **13** may include a semiconductor layer doped with P-type impurities. The diffusion layer **13** and the core layer **12** may have opposite conductivity types. For example, a P-N junction may be formed between the diffusion layer **13** and the core layer **12**. The impurity concentration of the diffusion layer **13** may be higher than the impurity concentration of the core layer **12**. As such, the diffusion layer **13** is designated as a “P+” region as shown in FIG. 1. In some embodiments, the diffusion layer **13** may be formed by diffusing P-type impurities into the substrate **10** from the surface **102**. P-type impurities may include trivalent elements such as boron, aluminum and gallium. In some embodiments, a diffusion boundary **131** may be formed between the diffusion layer **13** and the core layer **12**. In some embodiments, the diffusion boundary **131** is a boundary between a high concentration P-type impurity (P+) region and an N-type impurity (N) region.

(20) In some embodiments, the diffusion layer **13** may include a plurality of diffusion regions **14** separated from each other. The diffusion regions **14** and the diffusion layer **13** may have opposite conductivity types. For example, a P-N junction may be formed between each of the diffusion regions **14** and the diffusion layer **13**. The impurity concentration of the diffusion regions **14** may be higher than the impurity concentration of the diffusion layer **13**. As such, the diffusion regions **14** are designated as “N+” regions as shown in FIG. 1. In some embodiments, the impurity

concentration of the diffusion regions **14** may be about ten times or more than ten times higher than the impurity concentration of the diffusion layer **13**.

(21) Diffusion regions **14** may be formed adjacent to surface **102**. For example, diffusion regions **14** may be partially exposed from the surface **102**. For example, the surface of each of the diffusion regions **14** may be coplanar with the surface **102** of the substrate **10**. In this embodiment, the diffusion regions **14** do not extend to the diffusion boundary **131**. In some embodiments, the distance (e.g., the shortest distance) between diffusion regions **14** and diffusion boundary **131** is at least greater than about 15 μm .

(22) The electrode layer **15** may be disposed on the surface **101**. For example, the electrode layer **15** may be disposed on the diffusion layer **11**. The electrode layer **15** may contact or cover a portion of the diffusion layer **11**. The electrode layer **15** may completely cover the surface **101**. The electrode layer **15** can be regarded as an electrical terminal of the semiconductor structure **1**. For example, the electrode layer **15** may be electrically connected to the electrical terminal of the semiconductor structure **1**.

(23) The electrode layer **16** may be disposed on the surface **102**. The substrate **10** may be between the electrode layer **15** and the electrode layer **16**. For example, the electrode layer **16** may be disposed on the diffusion layer **13**. The electrode layer **16** may contact or cover a portion of the diffusion layer **13**.

(24) In some embodiments, the width of the electrode layer **15** may be greater than the width of the electrode layer **16** when viewed from a cross-sectional view. In some embodiments, the total surface area of the electrode layer **15** may be greater than the total surface area of the electrode layer **16**.

(25) In some embodiments, the exposed surface of one or more diffusion regions **14** from the surface **102** may be covered by the electrode layer **16**. For example, the electrode layer **16** may contact or cover one or more diffusion regions **14**. In some embodiments, the diffusion layer **13** may be coupled to each of diffusion regions **14** through the electrode layer **16**. For example, the electrode layer **16** may be configured to short-circuit the diffusion layer **13** and each of diffusion regions **14**. The diffusion layer **13** and each of the diffusion regions **14** can be regarded as an electrical terminal of the semiconductor structure **1**. For example, the diffusion layer **13** and each of the diffusion regions **14** may be electrically connected to one electrical terminal of the semiconductor structure **1**. The electrode layer **15** may be electrically connected to another electrical terminal of the semiconductor structure **1**. Compared to the general BJT, the diffusion layer **13** and each of the diffusion region **14** can be short-circuited to each other without passing through other circuit elements, so as to achieve the effect of short-circuiting the base and the emitter. Once the base and the emitter is shorted, it can be used as the diode element in the RCD circuit.

(26) In some embodiments, the electrode layer **15** and the electrode layer **16** may comprise copper (Cu), gold (Au), silver (Ag), aluminum (Al), nickel (Ni), titanium (Ti), tungsten (W), Tin (Sn), or other metals or alloys. In some embodiments, the electrode layer **15** and the electrode layer **16** may comprise the same material (e.g., aluminum). In some embodiments, the electrode layer **15** and the electrode layer **16** may comprise different materials. For example, the electrode layer **15** comprises aluminum, and the electrode layer **16** comprises silver, nickel, or titanium.

(27) In some embodiments, the substrate **10** may include a groove ring **10r** recessed downward or inward from the surface **102**. The groove ring **10r** may be located at the edge of the substrate **10**. The groove ring **10r** may surround the electrode layer **16**. In some embodiments, the groove ring **10r** may have an arc surface when viewed in a cross-sectional view. In some embodiments, the groove ring **10r** may have a rectangular shape such as a square or a rectangle when viewed from a top view. The electrode layer **16** may be located in the rectangle. However, the semiconductor structure **1** of the present disclosure may have the groove ring **10r** in any suitable shapes.

(28) In some embodiments, the groove ring **10r** may extend from the surface **102** down beyond the

diffusion boundary **112** to achieve a higher breakdown voltage for the semiconductor structure. For example, the groove ring **10r** may extend to the diffusion layer **11**. A portion of the diffusion layer **11** may be exposed from the groove ring **10r**. For example, the bottom of the groove ring **10r** may be located in the diffusion layer **11**. In some embodiments, the thickness H2 of groove ring **10r** (e.g., the longest distance from surface **102** to the bottom of groove ring **10r**) may be about 80 μm . In some embodiments, the thickness H2 of the groove ring **10r** may be in a range from about 30% to about 50% of the thickness H1 of the substrate **10**.

(29) The polysilicon layer **17** may be disposed on the groove ring **10r**. The polysilicon layer **17** may be located at the edge of the substrate **10**. The polysilicon layer **17** may surround the electrode layer **16**. The polysilicon layer **17** may include semi-insulating polycrystalline-silicon (SIPOS). In some embodiments, the polysilicon layer **17** may contact or cover the surface of the groove ring **10r**. In some embodiments, the polysilicon layer **17** may completely contact or cover the surface of the groove ring **10r**. In some embodiments, the polysilicon layer **17** may contact or cover a portion of the diffusion layer **11**. In some embodiments, polysilicon layer **17** may contact or cover a portion of the core layer **12**. In some embodiments, the polysilicon layer **17** may contact or cover a portion of the diffusion layer **13**.

(30) In some embodiments, the polysilicon layer **17** may contact or cover a portion of surface **102**. In some embodiments, the polysilicon layer **17** may contact or cover a portion of one or more diffusion regions **14**. In some embodiments, the polysilicon layer **17** may extend from the groove ring **10r** toward the electrode layer **16** and contact an edge of the electrode layer **16**. In some embodiments, the polysilicon layer **17** and the electrode layer **16** may be coplanar. In some embodiments, the thickness of the polysilicon layer **17** may be greater than the thickness of the electrode layer **16**. In some embodiments, the thickness of the electrode layer **16** may be greater than the thickness of the polysilicon layer **17**.

(31) The glass layer **18** may be disposed over the groove ring **10r** and the polysilicon layer **17**. The polysilicon layer **17** may be located between the glass layer **18** and the groove ring **10r**. The glass layer **18** may be located at the edge of the substrate **10**. The glass layer **18** may surround the electrode layer **16**. In some embodiments, the glass layer **18** may fill the groove ring **10r**. In some embodiments, the glass layer **18** may fully fill the groove ring **10r**. In some embodiments, the glass layer **18** may contact or cover a portion of polysilicon layer **17**.

(32) In some embodiments, the glass layer **18** may cover a portion of the surface **102**. In some embodiments, the glass layer **18** may cover a portion of the diffusion regions **14**. In some embodiments, the glass layer **18** may cover a portion of the diffusion layer **13**. In some embodiments, the glass layer **18** may cover a portion of the core layer **12**. In some embodiments, the glass layer **18** may cover a portion of the diffusion layer **11**.

(33) In some embodiments, the glass layer **18** may be separated from the surface **102** by the polysilicon layer **17**. In some embodiments, the glass layer **18** may be separated from the diffusion regions **14** by the polysilicon layer **17**. In some embodiments, the glass layer **18** may be separated from the diffusion layer **13** by the polysilicon layer **17**. In some embodiments, the glass layer **18** may be separated from the core layer **12** by the polysilicon layer **17**. In some embodiments, the glass layer **18** may be separated from the diffusion layer **11** by the polysilicon layer **17**. In some embodiments, the glass layer **18** may not contact the electrode layer **16**.

(34) Referring to FIG. 2, FIG. 2 shows a cross-sectional view of a semiconductor structure **2** in accordance with various embodiments of the present disclosure. The same or similar elements of the semiconductor structure **2** shown in FIG. 2 and the semiconductor structure **1** shown in FIG. 1 are denoted by the same reference numerals, and the detailed description of the elements will not be repeated.

(35) The conductivity type of the semiconductor structure **2** may be opposite to that of the semiconductor structure **1**. The semiconductor structure **2** may use a semiconductor substrate doped with P-type impurities. The core layer **12** may be a portion of the original P-type

monocrystalline silicon wafer. As such, the core layer **12** is marked as a “P” region in FIG. 2. The diffusion layer **11** may include a semiconductor layer doped with P-type impurities. The diffusion layer **11** and the core layer **12** may have the same conductivity type. However, the impurity concentration of the diffusion layer **11** may be higher than the impurity concentration of the core layer **12**. As such, the diffusion layer **11** is marked as a “P+” region in FIG. 2.

(36) The diffusion layer **13** may include a semiconductor layer doped with N-type impurities. The diffusion layer **13** and the core layer **12** may have opposite conductivity types. In some embodiments, a P-N junction may be formed between the diffusion layer **13** and the core layer **12**. The impurity concentration of the diffusion layer **13** may be higher than the impurity concentration of the core layer **12**. As such, the diffusion layer **13** is denoted as an “N+” region in FIG. 2.

(37) The diffusion layer **13** may include a plurality of diffusion regions **14** separated from each other. The diffusion regions **14** and the diffusion layer **13** may have opposite conductivity types. In some embodiments, a P-N junction may be formed between each of the diffusion regions **14** and the diffusion layer **13**. The impurity concentration of the diffusion regions **14** may be higher than the impurity concentration of the diffusion layer **13**. As such, the diffusion regions **14** are designated as “P+” regions in FIG. 2.

(38) In some embodiments, the thickness H1 of the substrate **10** of the semiconductor structure **2** may be in a range from about 240 μm to about 260 μm . In some embodiments, the thickness H2 of the groove ring **10r** (e.g., the longest distance from the surface **102** to the bottom of the groove ring **10r**) may be in a range from about 80 μm to about 120 μm . In some embodiments, the thickness H2 of the groove ring **10r** may be in a range from about 30% to about 50% of the thickness H1 of the substrate **10**.

(39) In some embodiments, the electrode layer **16** of the semiconductor structure **2** may contact or cover the polysilicon layer **17**. For example, a portion of the polysilicon layer **17** may be located between the surface **102** and the electrode layer **16** in a direction perpendicular to the surface **102**. For example, in the direction perpendicular to the surface **102**, the polysilicon layer **17** and the electrode layer **16** may partially overlap. In some embodiments, the electrode layer **16** of the semiconductor structure **2** may contact the glass layer **18**. However, in some embodiments, the electrode layer **16** of the semiconductor structure **2** may also be coplanar with the polysilicon layer **17** as shown in FIG. 1. Similarly, in some embodiments, the polysilicon layer **17** and the electrode layer **16** of the semiconductor structure **1** may partially overlap in the direction perpendicular to the surface **102** as shown in FIG. 2.

(40) Referring to FIGS. 3A-3C, FIGS. 3A-3C are top views of the semiconductor structures in accordance with various embodiments of the present disclosure. In some embodiments, the semiconductor structure **1** and the semiconductor structure **2** may have a top view shown in any of FIGS. 3A-3C.

(41) As shown in FIG. 3A, in some embodiments, diffusion regions (e.g., diffusion regions **14** of the semiconductor structure **1** and the semiconductor structure **2**) may include diffusion regions **14a** and **14b** that are separated from each other. In some embodiments, the diffusion regions including diffusion regions **14a** and **14b** may be arranged in a matrix. Although the drawings of the present disclosure depict a matrix having four columns and four rows, the semiconductor structures of the present disclosure may also have diffusion regions arranged in a matrix of any number of columns or rows.

(42) In some embodiments, the diffusion regions including diffusion regions **14a** and **14b** may be circular or quasi-circular (e.g., oval) in shape. However, the semiconductor structure of the present disclosure can also have diffusion regions of any shape.

(43) As shown in FIG. 3B, in some embodiments, diffusion regions (e.g., diffusion regions **14** of the semiconductor structure **1** and the semiconductor structure **2**) may include diffusion regions **14c**, **14d**, and **14e** that are separated from each other. The diffusion region **14e** may be located in the central region of the diffusion layer **13**. The diffusion region **14e** may be a rounded rectangle.

The diffusion region **14d** may surround the diffusion region **14e**. The diffusion region **14c** may surround the diffusion region **14d**. In some embodiments, the diffusion regions **14c**, **14d** and **14e** are concentric structures.

(44) FIG. **3C** is similar to FIG. **3B** except that the diffusion region **14e'** is located in the central region. As shown in FIG. **3C**, the diffusion region **14e'** is not solid.

(45) In some embodiments, the position, shape, area ratio, quantity, etc. of the diffusion regions can be adjusted so that the diffusion regions can be applicable to different circuits.

(46) Referring to FIGS. **4A-4H**, FIGS. **4A-4H** show one or more stages in a method of fabricating the semiconductor structure in accordance with various embodiments. At least some of these figures have been simplified to facilitate a better understanding of the primary aspects of the present disclosure.

(47) Referring to FIG. **4A**, the manufacturing method includes providing a substrate **10**. The substrate **10** may include a semiconductor substrate doped with N-type or P-type impurities. In the embodiments shown in FIGS. **4A** to **4I** and the following description, an N-type silicon wafer is used as an example. In some embodiments, the thickness of the substrate **10** may be in a range from about 350 μm to about 370 μm . In some embodiments, N-type silicon wafers with volume resistivity in a range from about 30 Ohm-cm to about 45 Ohm-cm may be used. The substrate **10** may have a surface **101** and a surface **102'** opposite to the surface **101**.

(48) Referring to FIG. **4B**, N-type impurities are diffused into the substrate **10** from the surface **101** and the surface **102'** of the substrate **10** using a double-sided diffusion technique. The double-sided diffusion may include an N-type impurity pre-deposition step and an N-type impurity main diffusion step. Phosphorus is selected as an example to illustrate this technique. Phosphorus oxychloride (POCl_3) can be used as the phosphorus source for the pre-deposition step. The pre-deposited substrate **10** is then subjected to the main diffusion step to advance the diffusion boundary and form a deeper N⁺ layer. The substrate **10** is double-diffused to form an N⁺/N/N⁺ structure including a diffusion layer **40** adjacent to the surface **102'**, a core layer **12** and a diffusion layer **11** adjacent to the surface **101**. In some embodiments, the thickness (or diffusion depth) H_3 of the diffusion layer **40** and the diffusion layer **11** may be about 120 μm .

(49) If a P-type silicon wafer is used as the substrate **10**, P-type impurities can be diffused into the substrate **10** from the surface **101** and the surface **102'** of the substrate **10**. The double-sided diffusion may include a P-type impurity pre-deposition step and a P-type impurity main diffusion step. Boron is selected as an example to illustrate this technique. For example, boron trichloride (BCl_3) can be used for the pre-deposition step as the source of boron. The pre-deposited substrate **10** is then subjected to a main diffusion step to advance the diffusion boundary and form a deeper P⁺ layer. The P⁺/P/P⁺ structure is formed after the substrate **10** is diffused on both sides.

(50) Referring to FIG. **4C**, a grinding process and a polishing process are performed on the substrate **10** from the surface **102'** to remove a portion of the diffusion layer **40**. As a result, the thickness H_4 of the diffusion layer **40** is reduced to about 20 μm . After the grinding process and the polishing process have been performed on the substrate **10**, the substrate **10** has a new surface **102**. After that, P-type impurities are diffused into the substrate **10** from the surface **102**. The diffusion process may include a P-type impurity pre-deposition step and a P-type impurity main diffusion step. Boron is selected as an example to illustrate this technique. For example, boron trichloride (BCl_3) can be used for the pre-deposition step as the source of boron. The pre-deposited substrate **10** is then subjected to the main diffusion step to invert the remaining diffusion layer **40** from N-type to P-type. After the substrate **10** is diffused, a P⁺/N/N⁺ structure is formed. The P⁺/N/N⁺ structure includes a diffusion layer **13** adjacent to the surface **102**, the core layer **12** and the diffusion layer **11** adjacent to the surface **101**. In some embodiments, the grinding process and the polishing process may be performed on the substrate **10** to completely remove the diffusion layer **40**. As a result, the core layer **12** is exposed. After that, P-type impurities are diffused into the core layer **12** to invert a portion of the core layer **12** from N-type to P-type.

(51) If a P-type silicon wafer is used as the substrate **10**, N-type impurities can be diffused from the surface **102** of the substrate **10** (diffusion layer **40** is partially removed) or from the core layer **12** (diffusion layer **40** is completely removed). The diffusion process may include an N-type impurity pre-deposition step and an N-type impurity main diffusion step. Phosphorus is selected as an example to illustrate this technique. For example, phosphorus oxychloride (POCl₃) can be used as the phosphorus source for the pre-deposition step. The main diffusion is then performed on the pre-deposited substrate **10** to invert the remaining diffusion layer **40** or a portion of the core layer **12** from P-type to N-type.

(52) Referring to FIG. 4D, a base mask (not shown) is formed on the surface **102** of the substrate **10**. The base mask may have a plurality of openings to define the diffusion regions **14** (the N⁺ regions **14**). The base mask can define the location of the P-N junction. For example, the base mask can define both the base and the emitter. After that, N-type impurities are diffused from the surface **102** of the substrate **10** to form a plurality of diffusion regions **14** separated from each other in the diffusion layer **13**. Phosphorus is selected as an example to illustrate this technique. Phosphorus oxychloride (POCl₃) can be used as the source of phosphorus. The base mask can block phosphorus from diffusing into the diffusion layer **13**. The base mask can be removed by etching, stripping, or other suitable processes.

(53) If a P-type silicon wafer is used as the substrate **10**, boron trichloride (BCl₃) can be used as a source of boron to form a plurality of P⁺ regions separated from each other in the diffusion layer **13**.

(54) Referring to FIG. 4E, a mesa mask (not shown) is formed on the surface **102** of the substrate **10**. The mesa mask may have openings to define the location of the groove ring **10r**. The mesa mask is used to form a mesa area surrounded by the groove ring **10r**. A chemically etched region is formed in the substrate **10** by photolithography techniques such as gluing, exposure and development. The chemically etched region is subjected to solution etching to form the groove ring **10r**. The groove ring **10r** may extend to the diffusion layer **11**. In some embodiments, the thickness H₂ of groove ring **10r** (e.g., the longest distance from surface **102** to the bottom of groove ring **10r**) may be about 80 μm. The mesa mask can be removed by etching, lift-off, or other suitable processes.

(55) Referring to FIG. 4F, a polysilicon layer **17** is formed over the groove ring **10r**. In some embodiments, the polysilicon layer **17** may be formed by a SIPOS deposition process.

(56) Referring to FIG. 4G, a glass layer is formed in the groove ring **10r** and on the polysilicon layer **17**. In some embodiments, the glass layer may be coated on substrate **10**. In some embodiments, a photoresist liquid may be incorporated into the glass liquid to form a photoresist liquid glass, which may be coated on the substrate **10**. In some embodiments, a glass mask (not shown) is formed on the surface **102** of the substrate **10**. The glass mask may have openings to define the location of the glass layer **18**. A portion of the glass layer may be removed, and the glass layer **18** may be formed using photolithography techniques and a firing process. After the portions of the glass layer have been removed, the polysilicon layer **17** on surface **102** is exposed from glass layer **18**. The glass mask can be removed by etching, lift-off, or other suitable processes.

(57) Referring to FIG. 4H, a contact mask (optionally, not shown in the figure) is formed on the polysilicon layer **17**. The contact mask may have openings to define the opening size of the polysilicon layer **17**. A portion of the polysilicon layer **17** exposed from the glass layer **18** on the surface **102** can be removed using the photolithography techniques to expose a portion of the diffusion layer **13** and a portion of the diffusion regions **14**. The exposed diffusion layer **13** and the exposed diffusion regions **14** will be in direct contact with a conductive contact structure formed thereon. The conductive contact structure is the electrode layer **16**. In this embodiment, a portion of the polysilicon layer **17** remains on the surface **102**. The contact mask can be removed by etching, lift-off or other suitable processes.

(58) In some embodiments, the opening size of the polysilicon layer **17** may be directly defined by

the glass layer **18** without using a contact mask. In other words, the polysilicon layer **17** exposed from the glass layer **18** on the surface **102** can be completely removed by the photolithography techniques.

(59) Referring to FIG. **4I**, the electrode layer **15** and the electrode layer **16** are formed on the surface **101** and the surface **102** through a metal mask (not shown in the figure), respectively. The diffusion layer **13** (equivalent to the base) and the diffusion region **14** (each of which is equivalent to the emitter) are shorted together. In this configuration, the substrate **10** is sandwiched between the electrode layer **15** and the electrode layer **16**. In some embodiments, the electrode layer **15** and the electrode layer **16** may be formed by sputtering, electroless plating, electroplating, printing, or other suitable processes. In some embodiments, the electrode layer **16** is not deposited on the polysilicon layer **17** (as shown in FIG. **1**). In some embodiments, the electrode layer **16** may be deposited on the polysilicon layer **17** (as shown in FIG. **2**). The metal mask can be removed by etching, lift-off, or other suitable processes. The semiconductor structure formed by the above steps can be the same as the semiconductor structure **1** shown in FIG. **1** or the semiconductor structure **2** shown in FIG. **2**.

(60) The manufacturing method of the semiconductor structure of the present application can realize the NPN or PNP transistor without using the existing BJT process, and during the forward conduction, the PN junction mainly plays the role of conducting current. During the turn-on and the turn-off processes, due to the impact of current and voltage, the NPN or PNP switch acts to reduce the oscillation in the circuit and improve the efficiency. Adjusting the ratio of the effective areas occupied by N⁺ or P⁺ according to different applications can make the structure suitable for different applications. As mentioned above, the fabrication method of the semiconductor structure of the present application uses four or five masks, including a base mask, a mesa mask, a glass mask, a contact mask (optionally) and a metal mask. Compared with the BJT manufacturing method (usually requiring at least six or seven masks), the present case requires fewer masks, so the manufacturing process is simplified and the cost is lower.

(61) Spatially relative terms such as “below,” “below,” “lower,” “above,” “upper,” “left,” “right,” etc. may be used herein for ease of description. The terms are used to describe the relationship between one component or feature and another component or feature in the drawings. In addition to the orientation depicted in the figures, spatially relative terms are intended to encompass different orientations of the device in use or operation. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted in a corresponding manner. It will be understood that when a component is referred to as being “connected” or “coupled” to another component, it can be directly connected or coupled to the other component or intervening components may be present.

(62) As used herein, the terms “substantially” and “about” are used to describe and explain small variations. When used in conjunction with an event or circumstance, the terms can refer to both instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs approximately. As used herein with respect to a given value or range, the term “about” generally means within $\pm 10\%$, $\pm 5\%$, $\pm 1\%$, or $\pm 0.5\%$ of the given value or range. A range may be expressed herein as one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints unless otherwise indicated. The term “substantially coplanar” may mean that two surfaces are positioned along the same plane within a few micrometers (μm), such as within $10\ \mu\text{m}$, $5\ \mu\text{m}$, $1\ \mu\text{m}$, or $0.5\ \mu\text{m}$ along the same plane. When values or properties are referred to as being “substantially” the same, the term can refer to values within $\pm 10\%$, $\pm 5\%$, $\pm 1\%$, or $\pm 0.5\%$ of the mean of the stated value.

(63) The foregoing has outlined features of several embodiments and detailed aspects of the present disclosure. The embodiments described in this disclosure may be readily utilized as a basis for designing or modifying other processes and structures for carrying out the same or similar purposes and/or achieving the same or similar advantages of the embodiments described herein. Such

equivalent constructions do not depart from the spirit and scope of the present disclosure, and without departing from the spirit and scope of the present disclosure, various changes, substitutions and alterations may be made.

(64) Although embodiments of the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims.

(65) Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

Claims

1. A method of manufacturing a semiconductor structure, comprising: providing a substrate having a first conductivity type, and having a first surface and a second surface opposite to the first surface; diffusing from the first surface of the substrate into the substrate to form a first diffusion layer having the first conductivity type; diffusing from the second surface of the substrate into the substrate to form a second diffusion layer having a second conductivity type; forming a plurality of diffusion regions having the first conductivity type in the second diffusion layer, wherein a first conductivity type impurity concentration of the plurality of diffusion regions is higher than a second conductivity type impurity concentration of the second diffusion layer; forming a first electrode layer on the first diffusion layer; forming a second electrode layer on the second diffusion layer, wherein the second electrode layer is in contact with the plurality of diffusion regions; and coupling the second diffusion layer to the plurality of diffusion regions through the second electrode layer, wherein, the substrate is between the first electrode layer and the second electrode layer.
 2. The method of claim 1, further comprising forming the second diffusion layer by diffusing from the second surface of the substrate into the substrate; further comprises: forming a diffusion layer having the first conductivity type; removing at least a portion of the diffusion layer; and inverting the rest of the diffusion layer from the first conductivity type to the second conductivity type through a diffusion process.
 3. The method of claim 1, further comprising: forming a square groove ring in the substrate; and forming a glass layer in the square groove ring, wherein the glass layer covers at least a portion of the second surface of the substrate.
 4. A method of manufacturing a semiconductor structure, comprising: forming a first diffusion layer on a first electrode layer; forming a core layer over the first diffusion layer; forming a second diffusion layer over the core layer; forming a plurality of diffusion regions in the second diffusion layer; forming a second electrode layer over the second diffusion layer and in contact with the plurality of diffusion regions; forming a square groove ring extending from the second electrode layer down to a diffusion boundary of the first diffusion layer; forming a glass layer over the square groove ring; and forming a polysilicon layer between the glass layer and the square groove ring.
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